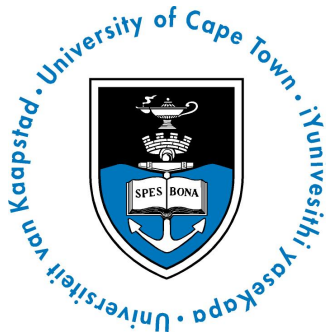


University of Cape Town



Supervised Learning Assignment 2

AirBnB Bookings Prediction

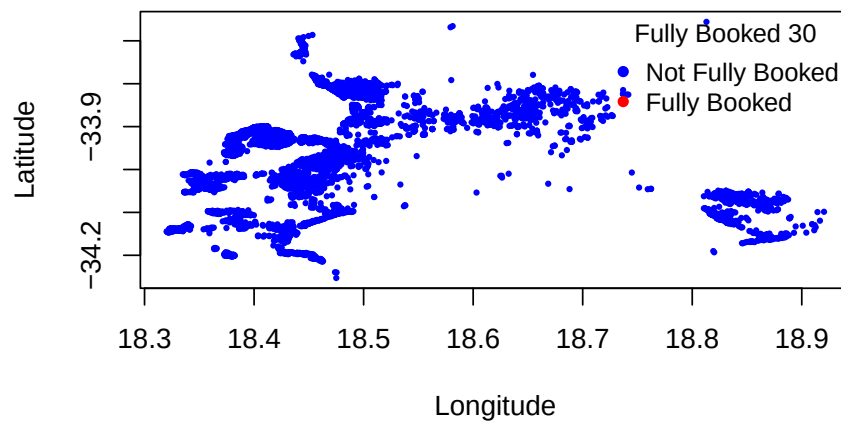
Pamela Afful

Student Number: AFFPAM001

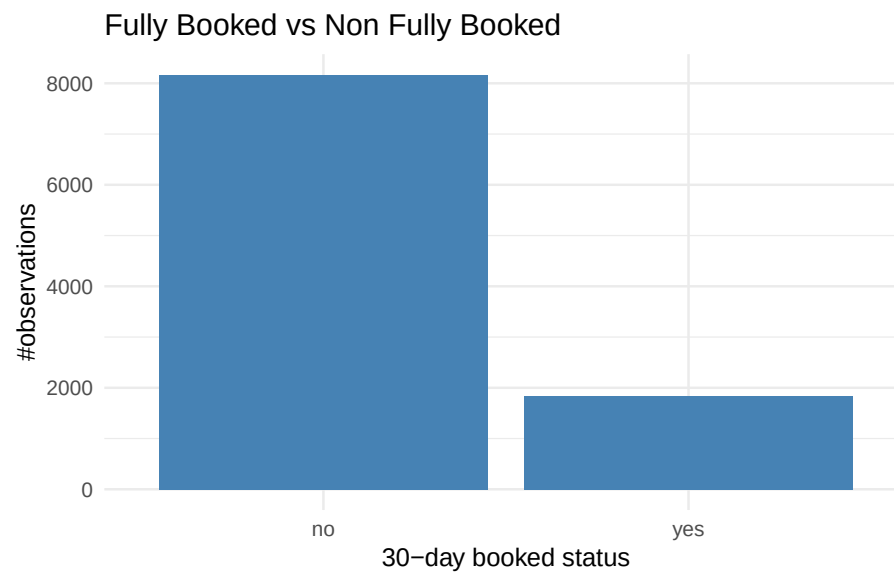
May 5, 2026

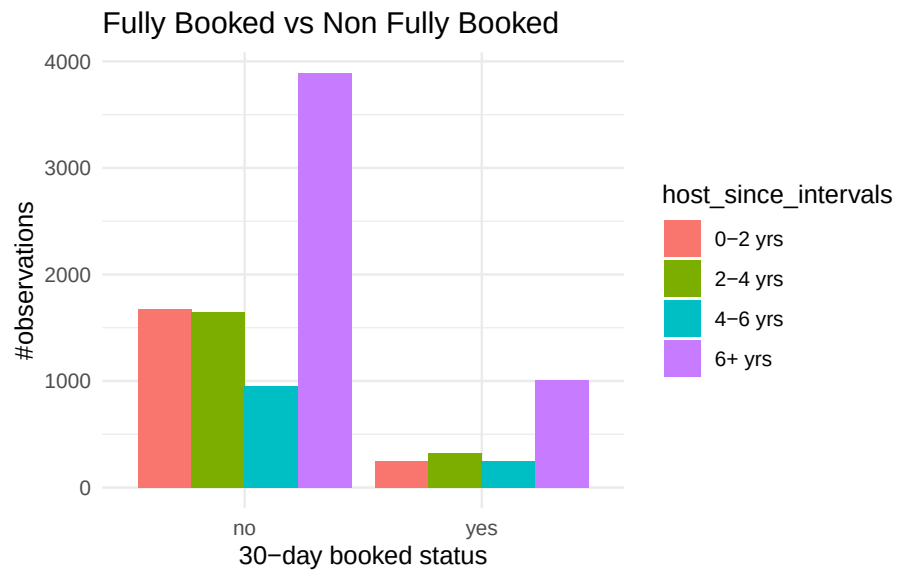
You can add options to executable code like this

Data Wrangling



EDA and Feature Selection





Use a novel decision tree to gauge which features could be important for determining the response variable. Since we have an imbalanced class, a weighted decision tree is used:

<https://machinelearningmastery.com/cost-sensitive-decision-trees-for-imbalanced-classification/>

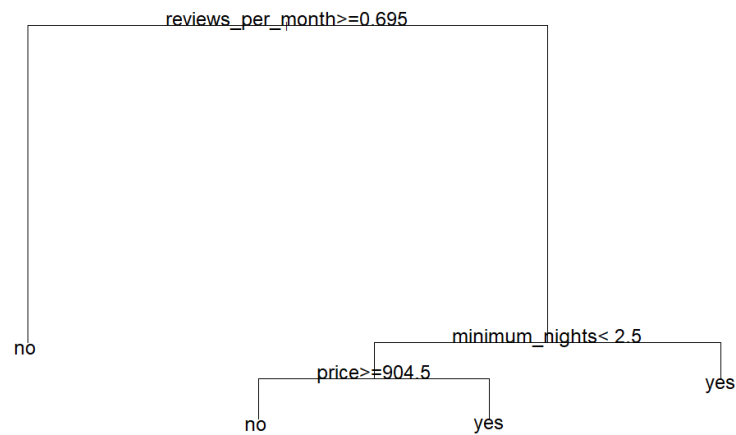
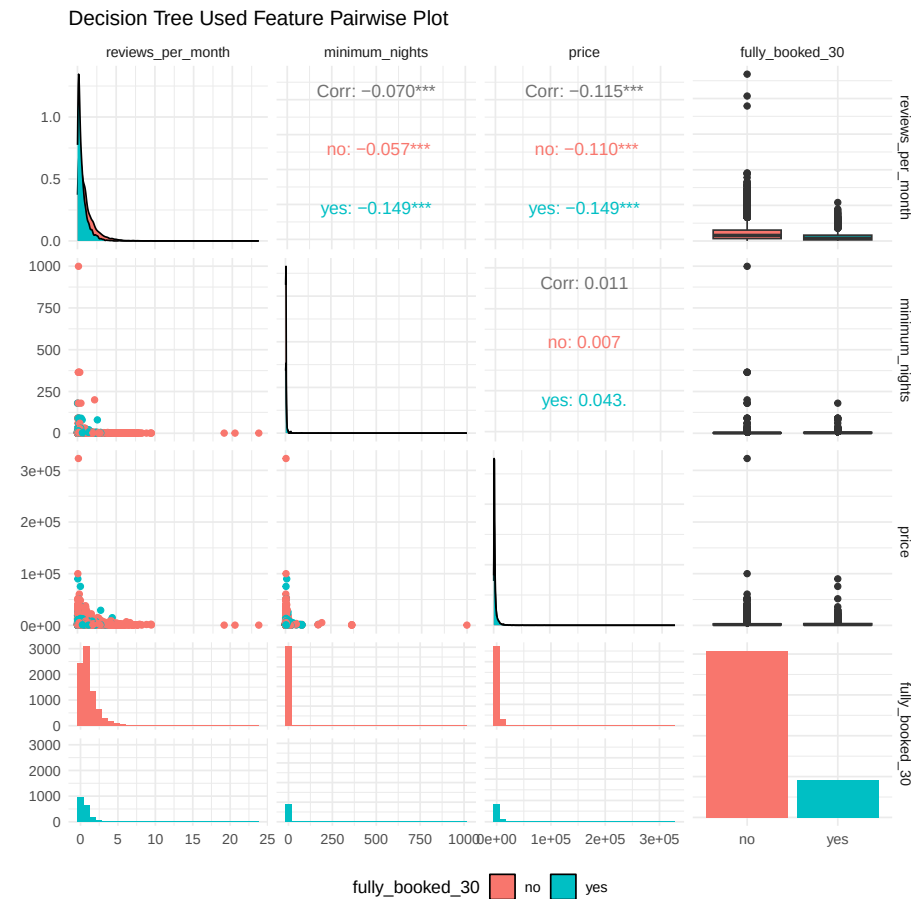


Figure 1: Initial Decision Tree Feature/ Class Splits

Features used to reduce RSS during splitting were **reviews_per_month** (the largest reduction), **minimum_night**, and **price**.

BnB listings with reviews_per_month greater than 0.695 all had terminal nodes with the categorisation “no” for the response variable. Price and minimum nights were determining factors for listings with a minimum number of nights less than 2.5, which ended in “yes”. Further splitting this region, for listings where the minimum was greater than 2.5, listings with prices greater than 905.5 were also categorised as “yes”.



- comeback and add more

Model Training

Baseline Model Vanilla Logistic Regression

Characteristic	log(OR) ^f	SE
(Intercept)	26*	10.2
host_response_time		
within an hour	—	—
within a few hours	0.18*	0.073
within a day	0.42***	0.097
a few days or more	0.47	0.324
host_response_rate	0.00	0.003
host_acceptance_rate	0.00*	0.002
host_is_superhost		
t	—	—
f	0.07	0.060
host_listings_count	-0.01***	0.001
host_verifications		
phone_only	—	—
phone_email	0.05	0.097
email_only	2.0	1.50
latitude	0.56*	0.278
longitude	-0.54*	0.252
room_type		
Entire home/apt	—	—
Private room	-0.15	0.103
accommodates	-0.01	0.030
bathrooms	-0.08	0.047
bedrooms	0.12*	0.056
beds	0.00	0.027
price	0.00	0.000
minimum_nights	-0.01	0.011
maximum_nights	0.00	0.000
minimum_nights_avg_ntm	0.02	0.011
maximum_nights_avg_ntm	0.00	0.000
number_of_reviews	0.00	0.001
reviews_per_month	-0.67***	0.054
review_scores_rating	0.23**	0.085
instant_bookable		
t	—	—
f	0.21**	0.073
host_since_months_int	0.00***	0.001
^f *p<0.05; **p<0.01; ***p<0.001		
Abbreviations: CI = Confidence Interval, OR = Odds Ratio, SE = Standard Error		
No. Obs. = 10,000		

Figure 2: Baseline Logistic Regression Summary

-
- Significant Features:
 - Intercept (which absorbes the omitted categorical variables: within a few hours, instant bookable=T)
 - Latitude and Longitude,

- Bedrooms,
- Reviews per month
- Review Score Ratings
- Instant bookable=F,
- Host Since (Measured in months)
- Insignificant Features:
 - The rest* comeback
 - Most interestingly the price!-

```
library(plotmo) #Specifically for glmnet plotting (also has a gbm function)

x_train <- select(heart, -DEATH_EVENT) |> slice(train)

# Fit the lasso and plot using plotmo
heart_l1 <- glmnet(x_train, y_train, alpha = 1, standardize = T, family = 'binomial')
plot_glmnet(heart_l1, xvar = 'norm')
```